

## Title

# THERMAL ABLATION VERSUS PARATHYROIDECTOMY FOR PRIMARY HYPERPARATHYROIDISM: A SYSTEMATIC REVIEW AND META-ANALYSIS

## Authors and Affiliation

**Lama Younes<sup>1</sup>, Zeinab Issa<sup>1</sup>, Aya Saadeddine<sup>2</sup>, Ghada El Hajj Fuleihan<sup>3</sup>, Marlene Chakhtoura<sup>3</sup>**

1. Department of Internal Medicine, American University of Beirut Medical Center, Beirut, Lebanon 2. Beirut Arab University, Faculty of Medicine

3. Calcium Metabolism and Osteoporosis Program, WHO Collaborating Center, American University of Beirut-Medical Center (AUBMC).

## Background

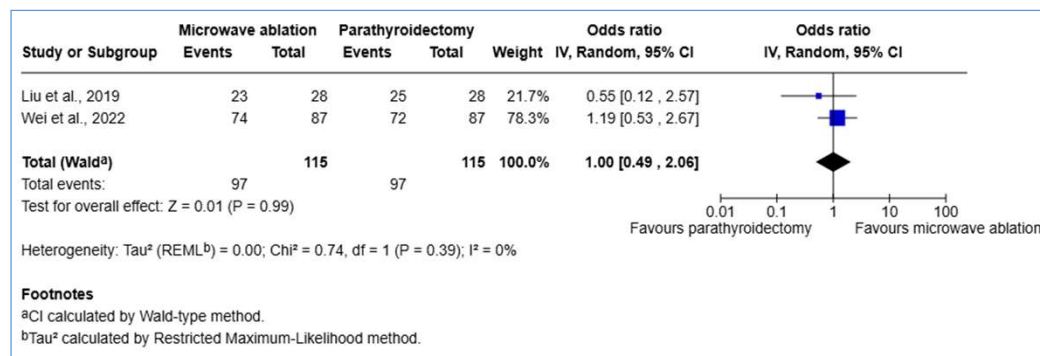
- Primary hyperparathyroidism (pHPT) is an endocrine disorder characterized by autonomous production of parathyroid hormone (PTH), most commonly due to a benign parathyroid adenoma. pHPT precipitates hypercalcemia, leading to nephrolithiasis, osteopenia/osteoporosis, gastrointestinal distress and neuropsychiatric changes.
- Parathyroidectomy (PTX) is the standard treatment for pHPT. Thermal ablation (TA) has emerged as a less invasive alternative.
- Our objective was to **compare the effectiveness and safety of thermal ablation versus PTX in adults with pHPT, assessing PTH, calcium and phosphorus levels, cure rates, and adverse effects.**

## Materials and Methods

- We conducted a comprehensive literature search of MEDLINE, EMBASE, and Web of science from inception to March 2025, following the PRISMA guidelines (PROSPERO protocol CRD420251009401).
- We included comparative studies reporting outcomes for adults with pHPT treated with thermal ablation and PTX. Cure was defined as normalization of serum calcium and PTH levels at 6 months follow up.
- Two reviewers screened, extracted data, and assessed risk of bias using ROBINS-I, independently and in duplicate.
- We performed random-effects meta-analysis (RevMan V6) when  $\geq 2$  studies with comparable interventions and outcome definitions were available. Otherwise, results were synthesized narratively.

## Results

- We screened 3,765 records at title/abstract level, assessed 11 full texts, and included 3 comparative retrospective cohort studies conducted in China.
- Mean age range: 53.5-57.7 years, proportion of females range: 65-79%.
- Preoperative serum calcium range: 2.69-2.80 mmol/L, reported adenoma size range: 1.4-2.13 cm.
- One study compared radiofrequency ablation (RFA) vs PTX<sup>1</sup>; reporting comparable clinical cure and safety outcomes.
- Meta-analysis was performed for microwave ablation (MWA) vs PTX<sup>2,3</sup> (2 studies; MWA n=115, PTX n=115). Cure rate, defined as normalization of serum calcium and PTH at 6 months follow-up, was similar between MWA and PTX (odds ratio [OR] 1.00, 95% CI 0.49–2.06,  $I^2=0\%$ ), Figure 1.
- Post-treatment calcium (mean difference [MD]  $-0.00$  mmol/L, 95% CI  $-0.20$  to  $0.19$ ,  $I^2=93\%$ ) and phosphorus (MD  $0.00$  mmol/L, 95% CI  $-0.05$  to  $0.05$ ,  $I^2=0\%$ ) were similar. Hypocalcemia was less frequent after MWA (OR 0.33, 95% CI  $0.12-0.92$ ,  $I^2=24\%$ ).
- Hoarseness showed a non-significant trend toward being more common with MWA (OR 3.54, 95% CI  $0.84-14.94$ ,  $I^2=0\%$ ).
- PTH levels could not be pooled because one study did not report SD.
- The three studies were judged to be at serious risk of bias, primarily due to confounding and participant selection.



## Conclusion

- In adults with pHPT, limited comparative evidence suggests MWA achieves **similar cure rates** to PTX with a significantly **reduced risk of hypocalcemia**, but conclusions are constrained by small study numbers and serious risk of bias.

## Impact Statement

- Demonstrating comparable cure rates to surgery and potentially reduced hypocalcemia introduces new, less invasive options for pHPT patients who refuse or are poor candidates for surgery.
- Larger prospective comparative studies are needed, including direct comparisons across ablation modalities.

## References:

1. Chai HH, Dai ZJ, Xu B, Hu QH, He HF, Xin Y, et al. Clinical and Economic Evaluation of Ultrasound-Guided Radiofrequency Ablation vs. Parathyroidectomy for Patients with Primary Hyperparathyroidism: A Cohort Study. Academic Radiology. 2023;30(11):2647-56.
2. Liu F, Yu X, Liu Z, Qiao Z, Dou J, Cheng Z, et al. Comparison of ultrasound-guided percutaneous microwave ablation and parathyroidectomy for primary hyperparathyroidism. International Journal of Hyperthermia. 2019;36(1):835-40.
3. Wei Y, Zhao ZL, Cao XJ, Peng LL, Li Y, Wu J, et al. Microwave ablation versus parathyroidectomy for the treatment of primary hyperparathyroidism: a cohort study. European Radiology. 2022;32(9):5821-30.